Bayesian Hierarchical Modeling of Cancer Mutation Rates

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Repository: github.com/annmariet/STAT405-project

1 Introduction

Somatic mutation rates, or genetic changes that occur in cells during a person's lifetime, for cancer-driving genes vary dramatically across tumor types. For example, the gene *TP53* is mutated in roughly 83% of lung squamous carcinomas but under 1% of thyroid cancers, while *EGFR* is a near-defining feature of lung adenocarcinoma and essentially absent elsewhere. Reporting each (gene, cancer-type) rate independently ignores the fact that most oncogenes (which promote cell growth when activated) and tumor suppressors have cross-cancer behaviour that could be used to improve estimates, especially for cancer types with limited sample sizes.

We ask: *how much does mutation prevalence vary across cancer types within a given gene, and what common gene-level structure can we recover by borrowing information across cancers?* Mutation prevalence refers to the proportion of tumors in a given cancer type that contain a mutation in a specific gene. The main modeling challenge is to balance information sharing across cancer types with the need to preserve gene-specific differences.

We take a Bayesian hierarchical approach on counts from 10 well-studied oncogenes and tumor suppressors across 20 TCGA PanCancer Atlas cancer types (200 (gene, cancer) observations). Our hierarchical Beta-Binomial model yields a per-gene concentration parameter s_g that directly quantifies cross-cancer consistency, a summary that neither independent Beta-Binomials nor fully pooled rates produce.

2 Literature Review

Pan-cancer cancer genomics has shown that somatic mutation patterns vary substantially across tumor types and are strongly structured by tissue and cell of origin. The TCGA Pan-Cancer Analysis Project established the value of analyzing tumors jointly across cancer types to identify both shared and specific genomic features (Weinstein et al. 2013). Later analyses of roughly 10,000 tumors across 33 cancers found that molecular clustering is organized primarily by histology, tissue type, and anatomic origin (Hoadley et al. 2018). These findings suggest that mutation prevalence should not be treated as either fully pooled across cancers or fully independent, but instead, models should allow for information sharing across cancer types.

A challenge in modeling mutation data is that mutation rates are not directly comparable across genes or cancer types. Lawrence et al. (2013) showed that the identification of genes associated with

cancers is complicated by mutational differences arising from biological and technical sources. This means that comparisons of mutation rates can be misleading. While this work focuses on detecting significantly mutated genes, it motivates our question of how to model mutation prevalence in a way that accounts for variability without discarding shared information.

Bayesian and hierarchical models are a natural way to address this problem and have been used in several cancer genomics studies. In a recent large-scale study, Scharpf et al. (2022) used Bayesian hierarchical models to estimate mutation prevalence and co-mutation patterns for RAS genes across multiple cancer types, showing that both prevalence and co-occurrence patterns depend strongly on tumor context. Similarly, Samorodnitsky, Hoadley, and Lock (2020) developed a Bayesian hierarchical model for the effect of somatic mutations on survival, motivated by the idea that mutation effects may be similar, but not identical, across cancer types. These studies support the use of hierarchical Bayesian modeling in pan-cancer studies, but they focus on either co-mutation structure or survival outcomes rather than direct estimation of cross-cancer mutation prevalence within genes.

We build on well-established hierarchical Bayesian methods in a simplified and interpretable setting. By modeling mutation counts using a hierarchical Beta-Binomial structure and focusing on cross-cancer variation within genes, the model is a middle ground between independent estimates and interaction-based approaches.

3 Data and Models

3.1 Data

Mutation counts were obtained from the TCGA PanCancer Atlas (Weinstein et al. 2013; Hoadley et al. 2018) via the cBioPortal API (Cerami et al. 2012) using the `cbioportalR` R package. For each of $G = 10$ genes (TP53, KRAS, PIK3CA, BRAF, PTEN, APC, ARID1A, EGFR, IDH1, CDKN2A) and $C = 20$ cancer types with at least 150 sequenced tumors, we computed y_{gc}, the number of tumors with a mutation in gene g in cancer type c, and n_{gc}, the total number of sequenced tumors for that cancer type to give us y_{gc}/n_{gc}, the observed mutation prevalence. This gives $N = G \times C = 200$ observations. Observed rates range from 0.004 (TP53 in thyroid) to 0.83 (TP53 in lung squamous).

3.2 Model 1: Independent Beta-Binomial (baseline)

A conjugate baseline assumes every (gene, cancer) pair is independent:

$$y_{gc} \mid \theta_{gc} \sim \text{Binomial}(n_{gc}, \theta_{gc}) \tag{1}$$

$$\theta_{gc} \sim \text{Beta}(1, 1). \tag{2}$$

This returns a $\text{Beta}(1 + y_{gc},\, 1 + n_{gc} - y_{gc})$ posterior per cell and cannot share information across cancer types.

3.3 Model 2: Hierarchical Beta-Binomial (partial pooling)

We pool within each gene by placing a gene-level prior on the rates:

$$y_{gc} \mid \theta_{gc} \sim \text{Binomial}(n_{gc}, \theta_{gc}) \tag{3}$$

$$\theta_{gc} \mid \mu_g, s_g \sim \text{Beta}(\mu_g s_g,\, (1 - \mu_g) s_g) \tag{4}$$

$$\mu_g \sim \text{Beta}(1, 1), \quad s_g \sim \text{Exponential}(0.1). \tag{5}$$

The (μ_g, s_g) parameterization is interpretable: μ_g is gene g's typical rate across cancer types and s_g is its concentration. Large s_g means the gene behaves consistently across tissues, small s_g means heterogeneity. The Exp(0.1) prior on s_g has mean 10 and is weakly informative, placing substantial mass over a wide range of concentration values. This allows both weak ($s \approx 2$) and strong pooling ($s \gg 50$), while still regularizing extreme values.

Identifiability note: with 20 observations per gene, per-gene hyperparameters are well-identified. We do not add a further hyper-prior pooling across genes, as 10 genes provides limited information for a second hierarchical level (Gelman et al. 2013, ch. 5).

4 Posterior Computation

We compute the hierarchical posterior with two methods: Hamiltonian Monte Carlo (HMC) and automatic differentiation variational inference (ADVI) (Kucukelbir et al. 2017) as a fast approximation. Both use Stan (via `cmdstanr`) with the same model code (Appendix A). To ensure correctness, both methods were implemented using the same Stan model and validated using standard MCMC diagnostics and posterior predictive checks.

HMC. Four chains, 1000 warmup + 1000 sampling iterations each, `adapt_delta = 0.95`, `max_treedepth = 12`. Sampling completed in 12 seconds with zero divergent transitions, maximum $\hat{R} = 1.005$, and minimum bulk ESS of 2,085 across all 220 parameters. Diagnostics pass comfortably (Appendix B).

ADVI. Two variational families are available in Stan: *mean-field* (fully factorized Gaussian on unconstrained parameters) and *full-rank* (multivariate Gaussian with learned covariance). We fit both with the default ELBO convergence criterion.

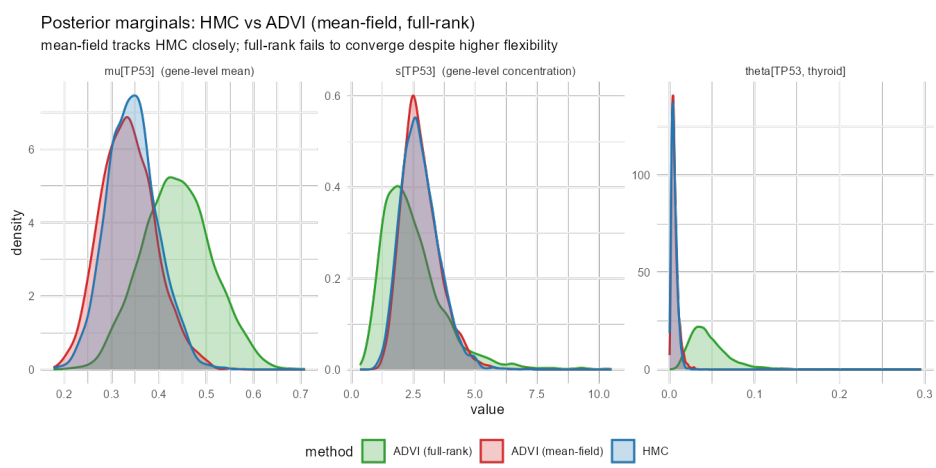

Figure 1: Posterior marginals for three parameters under HMC, mean-field ADVI, and full-rank ADVI. Mean-field tracks HMC closely on all three. Full-rank shifts μ_{TP53} by nearly 2σ and places most of $\theta_{\text{TP53,thyroid}}$ mass off the HMC posterior.

We assess approximation quality by comparing posterior marginals (Figure 1). For this hierarchical Beta-Binomial model, mean-field ADVI closely matches HMC for both hyperparameters and a representative θ_{gc}. In contrast, full-rank ADVI performs poorly despite using a more flexible

variational family and substantially more optimization steps (21,000 vs 700). Its posterior mean for μ_{TP53} is biased upward by 0.09, and its marginal for $\theta_{\text{TP53,thyroid}}$ peaks at 0.047, while the HMC posterior is near 0.006.

These results suggest that the more flexible variational approximation is harder to optimize and can produce misleading results without a clear warning. Mean-field ADVI provides an approximation very similar to HMC and is much faster.

5 Results

5.1 Hierarchical vs independent rates

Posterior predictive densities from the hierarchical model cover the observed rate distribution well (Figure 2), and leave-one-out cross-validation [via the `loo` package] favours the hierarchical model by an ELPD difference of -28.0 (SE $= 5.4$) over the independent baseline. This provides evidence that borrowing information across cancer types improves the predictive performance of our model.

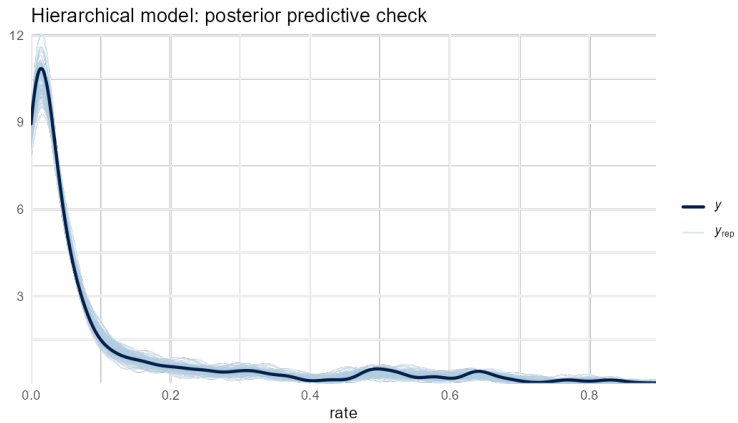

Figure 2: Hierarchical model posterior predictive check: observed rate density (dark) against 100 replicated datasets (light).

5.2 Gene-level structure

We assess the gene-level posterior predictive distribution, $\text{Beta}(\mu_g s_g, (1-\mu_g)s_g)$ marginalized over the posterior of (μ_g, s_g). This is the mutation rate we would expect for unseen cancer types for gene g (Figure 3). Three patterns emerge across genes:

1. *Broadly active tumor suppressors* (TP53): wide predictive mass across $[0, 0.9]$ with posterior mean $\mu_{\text{TP53}} = 0.34$ and low concentration $s_{\text{TP53}} = 2.8$. This reflects the role of TP53 as a general tumor suppressor that is important in many cancers, but to different degrees.
2. *Cancer-specific oncogenes* (EGFR, CDKN2A): mass concentrated near zero with rare high-rate outliers. $\mu_{\text{EGFR}} = 0.044$, $s_{\text{EGFR}} = 18.5$ indicate that EFGR mutations are rare in most cancer types but occur in high concentration in a small number of specific cancers, mostly lung adenocarcinoma in this dataset.
3. *Intermediate* (PIK3CA, KRAS, PTEN): posterior means in the 0.09–0.13 range with moderate s_g. These genes are involved in multiple cancer types but do not have the same spread of TP53 or concentration of EGFR or CDKN2A.

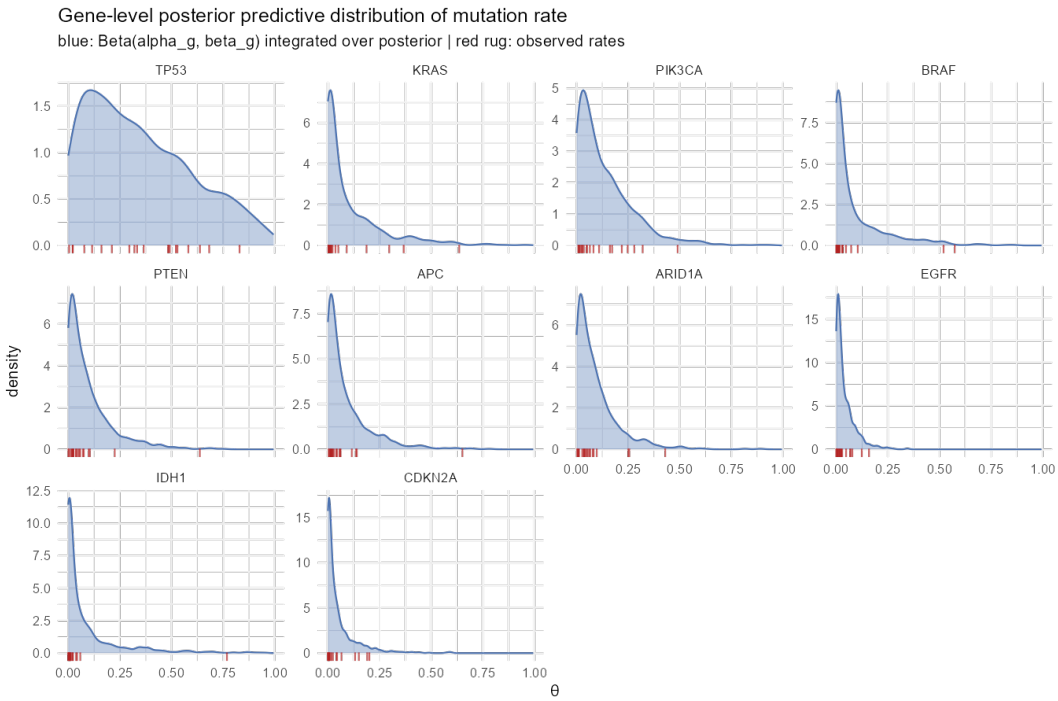

Figure 3: Gene-level posterior predictive distribution of θ (blue density), overlaid with observed rates as a red rug. TP53 spans a wide range; EGFR and CDKN2A are concentrated near zero with outliers.

5.3 Prior sensitivity

We refit the hierarchical model under two alternative priors on s_g. A tighter Gamma$(2, 0.1)$ (mean 20) and diffuse half-Cauchy$(0, 25)$ (Figure (**ref?**)(fig:prior-sens)). These priors all reflect different assumptions about the degree of cross-cancer consistency. The Gamma prior concentrates mass on moderate values of s_g, favouring stronger pooling across cancer types, while the half-Cauchy prior is heavy tailed which allows for the possibility of very large s_g. Across all 10 genes and three priors, the maximum shift in $\hat{\mu}_g$ is 0.010, while s_g estimates shift by 10–20%, most visibly for genes whose concentration is driven by a single dominant cancer (e.g., EGFR: 18.5, 21.1, 22.0 under Exponential, Gamma, and Cauchy). Gene rankings by μ_g are identical under every prior (Appendix C), confirming that cross-cancer mean rates are data-driven and that prior choice affects only how strongly the model interprets heterogeneity around those means.

6 Discussion

Our hierarchical Beta-Binomial quantifies how mutation prevalence varies across cancer types within a gene. The gene-level concentration parameter s_g provides a clear summary of cross-cancer mutation consistency, separating broadly-active tumor suppressors (TP53, $s \approx 3$) from narrowly cancer-specific oncogenes (EGFR, $s \approx 19$). These patterns align with known biological behavior of these genes, suggesting that the model is capturing meaningful structure in the data.

The model further provides a useful predictive framework. The gene-level posterior predictive distribution can be used to estimate mutation prevalence in a new or less-studied cancer type. This

may be helpful for guiding expectations in small datasets or early-stage clinical studies.

We have two main methodological takeaways. First, while HMC provides reliable posterior inference, mean-field ADVI offers a much faster approximation that performed similarly to HMC. However, full-rank ADVI did not align well with HMC, despite being more flexible. This suggests that increased model flexibility does not always lead to better approximations.

Overall, this analysis demonstrates how hierarchical Bayesian modeling can be used to estimate mutation prevalence from real-world cancer data. By balancing infomration sharing across cancer types with gene-specific differences, the model provides improved estimation and summaries of cross-cancer mutation behaviour.

6.1 Limitations

Our model pools information within gene, but does not extend this structure across genes. A second hierarchical level ($\mu_g \sim \text{Beta}(A, B)$ with hyper-hyperpriors) is natural but borderline-identifiable with only 10 genes. We also treat cancer types as independent, ignoring the tissue-of-origin structure that has been shown to be a dominant source of molecular variation in pan-cancer analyses (Hoadley et al. 2018). In addition, mutation status is treated as binary which may over-simplify differences between mutation types and locations (e.g. *KRAS* G12C vs G12D). Finally, for simplicity, our model assumes independence across genes and therefore does not capture known patterns of co-occurrence and mutual exclusivity in cancer genomics, which have been linked to underlying pathway interactions (El Tekle et al. 2021; Unni et al. 2015), and can be modeled using more complex approaches such as Bayesian networks (Kuipers et al. 2018). These limitations suggest clear directions for further developing this analysis.

Team contributions

This project was done collaboratively, with idea generation, model selection, coding, literature reviewing, and writing split evenly between us. Due to some technical difficulties and a lot of in-person work, most of the commits on the Github repository are by Annmarie, however, this is not representative of the division of work and was just for convenience.

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A Stan model code

Model 1 (naive independent Beta-Binomial):

```
data {
  int<lower=1> N;
  array[N] int<lower=0> y;
  array[N] int<lower=1> n;
}
parameters {
  vector<lower=0, upper=1>[N] theta;
}
model {
  theta ~ beta(1, 1);
  y ~ binomial(n, theta);
}
generated quantities {
  array[N] int y_rep;
  vector[N] log_lik;
  for (i in 1:N) {
    y_rep[i]   = binomial_rng(n[i], theta[i]);
    log_lik[i] = binomial_lpmf(y[i] | n[i], theta[i]);
```

```
  }
}
```

Model 2 (hierarchical Beta-Binomial, primary priors):

```
data {
  int<lower=1> N;
  int<lower=1> G;
  array[N] int<lower=1, upper=G> gene;
  array[N] int<lower=0> y;
  array[N] int<lower=1> n;
}
parameters {
  vector<lower=0, upper=1>[G] mu;
  vector<lower=0>[G]          s;
  vector<lower=0, upper=1>[N] theta;
}
transformed parameters {
  vector<lower=0>[G] alpha_ = mu .* s;
  vector<lower=0>[G] beta_  = (1 - mu) .* s;
}
model {
  mu ~ beta(1, 1);
  s  ~ exponential(0.1);
  for (i in 1:N)
    theta[i] ~ beta(alpha_[gene[i]], beta_[gene[i]]);
  y ~ binomial(n, theta);
}
generated quantities {
  array[N] int y_rep;
  vector[N] log_lik;
  for (i in 1:N) {
    y_rep[i]   = binomial_rng(n[i], theta[i]);
    log_lik[i] = binomial_lpmf(y[i] | n[i], theta[i]);
  }
}
```

B Diagnostics

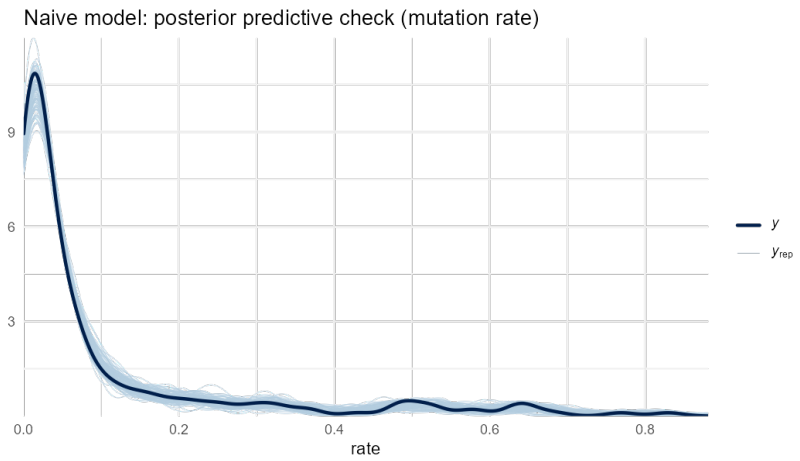

Figure 4: Naive model posterior predictive check.

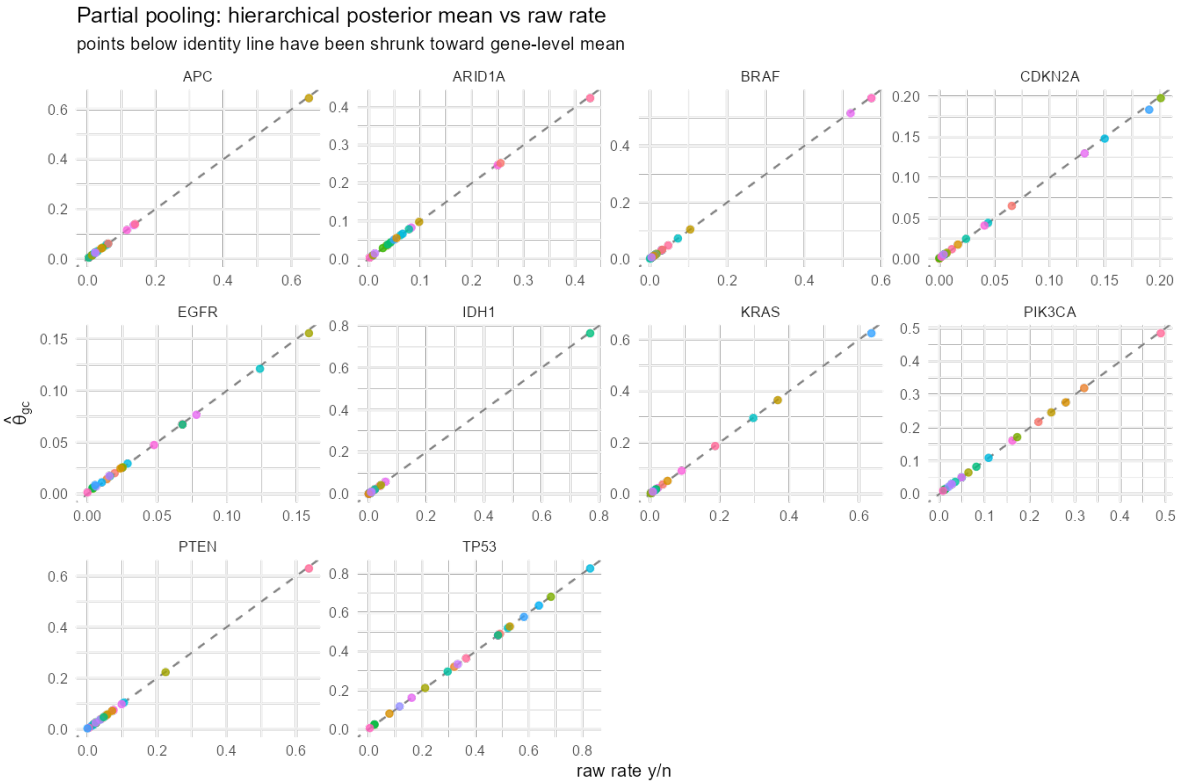

Figure 5: Hierarchical posterior mean $\hat{\theta}_{gc}$ vs raw rate y/n. With $n_{gc} \geq 150$ per cell, shrinkage is small in absolute terms; the hierarchical payoff is in the gene-level hyperparameters (see Fig. 2).

C Prior Evaluation

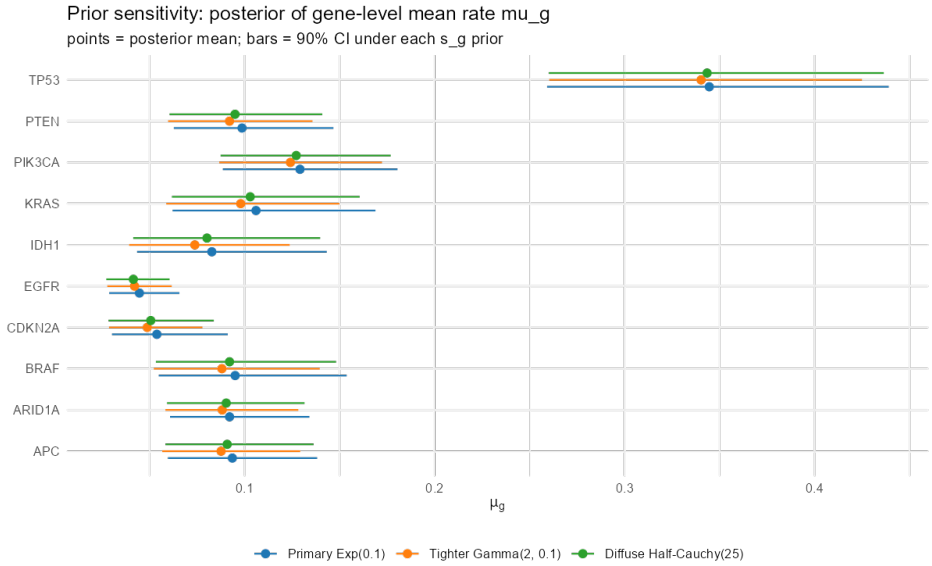

Figure 6: Posterior of gene-level mean rate μ_g under three priors on s_g. Point estimates are stable within sampling error; credible intervals widen slightly under the tighter prior for genes with low observed concentration.

D Reproduction

All analyses use R 4.x, `cmdstanr` 0.8+, and the `loo`, `bayesplot`, `posterior`, and `tidyverse` packages. The data pull is cached at `data/raw/cbioportal_pull_2026-04-14.csv`; the full pipeline is:

```
source("R/02_clean_data.R")        # build processed/mutation_counts.rds
source("R/03_fit_naive.R")         # Model 1 + PPC
source("R/04_fit_hier.R")          # Model 2 + shrinkage + LOO
source("R/04b_gene_predictive.R") # gene-level predictive figure
source("R/05_fit_advi.R")          # ADVI mean-field + full-rank
source("R/06_prior_sensitivity.R")# prior sweep
rmarkdown::render("report/report.Rmd")
```